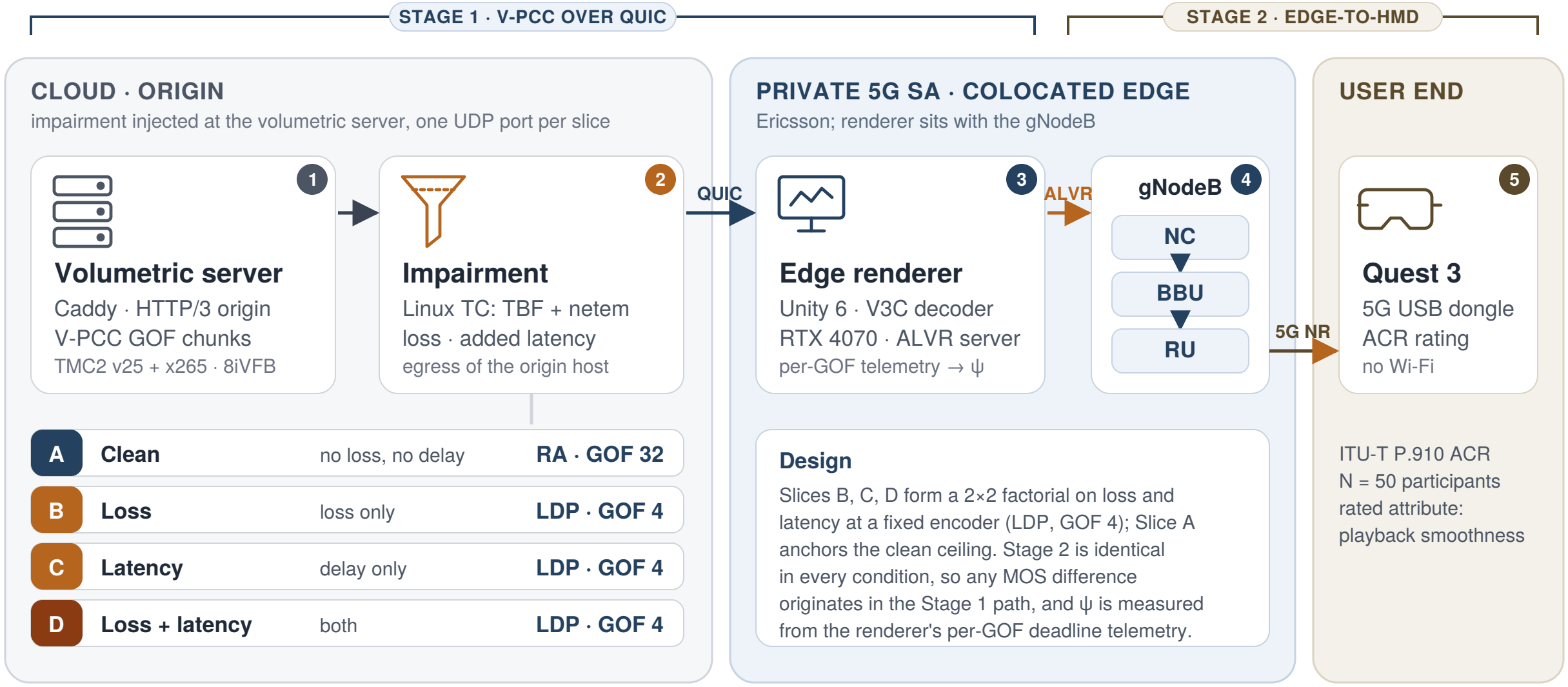
STAGE 1 · V-PCC OVER QUIC
STAGE 2 · EDGE-TO-HMD
CLOUD · ORIGIN
impairment injected at the volumetric server, one UDP port per slice
1
Volumetric server
Caddy · HTTP/3 origin
V-PCC GOF chunks
TMC2 v25 + x265 · 8iVFB
2
Impairment
Linux TC: TBF + netem
loss · added latency
egress of the origin host
QUIC
A Clean no loss, no delay RA · GOF 32
B Loss loss only LDP · GOF 4
C Latency delay only LDP · GOF 4
D Loss + latency both LDP · GOF 4
PRIVATE 5G SA · COLOCATED EDGE
Ericsson; renderer sits with the gNodeB
3
Edge renderer
Unity 6 · V3C decoder
RTX 4070 · ALVR server
per-GOF telemetry → ψ
ALVR
gNodeB
4
NC
BBU
RU
Design
Slices B, C, D form a 2×2 factorial on loss and latency at a fixed encoder (LDP, GOF 4); Slice A anchors the clean ceiling. Stage 2 is identical in every condition, so any MOS difference originates in the Stage 1 path, and ψ is measured from the renderer's per-GOF deadline telemetry.
5G NR
USER END
5
Quest 3
5G USB dongle
ACR rating
no Wi-Fi
ITU-T P.910 ACR
N = 50 participants
rated attribute:
playback smoothness